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$$\left(\int_1^2 \frac{e^{\frac{1}{x}}}{x^2} dx = \int_1^2 e^{\frac{1}{x}-2} dx \right)$$

$$\int_1^2 \frac{e^{\frac{1}{x}}}{x^2} dx$$

$$e^{\frac{1}{x}} = u$$

$$du = -\frac{e^{\frac{1}{x}}}{x^2} dx$$

$$-\int_1^2 \left(-\frac{e^{\frac{1}{x}}}{x^2} \right) dx = -\int_1^2 1 du =$$

$$= -\int_1^2 1 du = -u \Big|_1^2 = -e^{\frac{1}{x}} \Big|_1^2 =$$

$$= (-e^{\frac{1}{2}}) - (-e) = -e^{\frac{1}{2}} + e$$

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$$\tan^{-1}(\tan \frac{3\pi}{4})$$

$$\tan^{-1}(x) \in (-\frac{\pi}{2}, \frac{\pi}{2})$$

$$\tan \frac{3\pi}{4} = \tan(\pi - \frac{\pi}{4}) =$$

$$= -\tan \frac{\pi}{4}$$

$$\tan^{-1}(-\tan \frac{\pi}{4}) = -\frac{\pi}{4}$$

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$$\tanh(\ln x) = \frac{x^2 - 1}{x^2 + 1} \text{ because:}$$

$$\sinh(x) = \frac{e^x - e^{-x}}{2}$$

$$\cosh(x) = \frac{e^x + e^{-x}}{2}$$

$$\tanh(x) = \frac{\sinh(x)}{\cosh(x)}$$

$$\tanh(x) = \frac{\frac{e^x - e^{-x}}{2}}{\frac{e^x + e^{-x}}{2}} =$$

$$= \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$\tanh(\ln x) = \frac{e^{\ln x} - e^{-\ln x}}{e^{\ln x} + e^{-\ln x}}$$

$$\tanh(\ln x) = \frac{x - \frac{1}{x}}{x + \frac{1}{x}} =$$

$$= \frac{\frac{x^2 - 1}{x}}{\frac{x^2 + 1}{x}} = \frac{x^2 - 1}{x^2 + 1}$$

$$\tanh(\ln x) = \frac{x^2 - 1}{x^2 + 1}$$

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$$\lim_{x \rightarrow 0^+} (\cos x)^{\frac{1}{x^2}} = y$$

$$\lim_{x \rightarrow 0^+} \ln(\cos x)^{\frac{1}{x^2}} = \ln y$$

$$\lim_{x \rightarrow 0^+} \frac{\ln(\cos x)}{x^2} = \ln y$$

$$\lim_{x \rightarrow 0} \frac{[\ln(\cos x)]'}{(x^2)'} = \frac{-\sin x}{\cos x \cdot 2x}$$

$$\lim_{x \rightarrow 0} \frac{(-\sin x)'}{(\cos x \cdot 2x)'} = \frac{-\cos x}{-\sin x \cdot 2x + 2\cos x}$$

$$\lim_{x \rightarrow 0} \frac{-\cos x}{2\cos x - (\sin x \cdot 2x)} = \frac{-1}{2 - 0} = -\frac{1}{2}$$

$$\ln y = -\frac{1}{2}$$

$$y = \lim_{x \rightarrow 0^+} (\cos x)^{\frac{1}{x^2}} = e^{-\frac{1}{2}}$$

$$\lim_{x \rightarrow 0} (\cos x)^{\frac{1}{x^2}} = e^{-\frac{1}{2}}$$